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Zhang; Tao et al.

CAMERA OPTICAL LENS

Abstract

The present disclosure relates to the field of optical lenses, and more specifically to a camera optical lens. The camera optical lens includes five lenses, which include, from an object side to an image side in sequence: a first lens having a positive refractive power, a second lens having a positive refractive power, a third lens having a negative refractive power, a fourth lens having a positive refractive power, and a fifth lens having a negative refractive power. The camera optical lens satisfies following relationships: $3.00 \leq f2/f \leq 12.00$; $1.10 \leq (R9+R10)/(R9-R10) \leq 1.90$; $1.00 \leq d1/d2 \leq 4.00$; and $2.00 \leq R6/R5 \leq 15.00$. The camera optical lens according to the present disclosure can meet the design requirement of large aperture, wide-angle and ultra-thinness.

Inventors: Zhang; Tao (Changzhou, CN), Zhou; Shunda (Changzhou, CN)
Applicant: AAC OPTICS (CHANGZHOU) CO., LTD. (Changzhou City, CN)
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of priority under the Paris Convention to Chinese Patent Application No. 202410219462.6, entitled "CAMERA OPTICAL LENS," filed on Feb. 28, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to optical lens, in particular to a camera optical lens suitable for handheld devices, such as smart phones and digital cameras, and imaging devices, such as monitors and PC.

BACKGROUND

[0003] With the emergence of smart phones in recent years, the demand for miniature camera lens is increasing day by day. Due to the reduction in pixel size of photosensitive devices, coupled with the current development trend of electronic products being that their functions should be better and their shape should be thin and small, miniature camera lens with good imaging quality therefor has become a mainstream in the market. In order to obtain better imaging quality, the lens adopts a multiple-piece lens structure. And, with the development of technology and the increase of the diverse demands of users, and under this circumstances that the pixel area of photosensitive devices is shrinking steadily and the requirement of the system for the imaging quality is improving constantly, the five-piece lens structure gradually appear in lens design. There is an urgent need for wide-angle camera lenses which have good optical characteristics, small volume, and the chromatic aberration of which is fully corrected.

SUMMARY

[0004] In view of the above, the present disclosure is intended to provide a camera optical lens, which has good optical characteristics and satisfies the design requirements of large aperture, ultra-thinness, and wide angle.
[0005] In order to achieve the above objective, a solution of the present disclosure provides a camera optical lens. The camera optical lens includes five lenses. The five lenses include, from an object side to an image side in sequence: a first lens having a positive refractive power, a second lens having a positive refractive power, a third lens having a negative refractive power, a fourth lens having a positive refractive power, and a fifth lens having a negative refractive power. The camera optical lens satisfies following relationships: $3.00 \leq f2/f \leq 12.00$; $1.10 \leq (R9+R10)/(R9-R10) \leq 1.90$; $1.00 \leq d1/d2 \leq 4.00$; and $2.00 \leq R6/R5 \leq 15.00$. Where, f represents a focal length of the camera optical lens, $f2$ represents a focal length of the second lens, $R9$ represents a central curvature radius of an object-side surface of the fifth lens, $R10$ represents a central curvature radius of an image-side surface of the fifth lens, $d1$ represents an on-axis thickness of the first lens, $d2$ represents an on-axis distance from an image-side surface of the first

lens to an object-side surface of the second lens, R5 represents a central curvature radius of an object-side surface of the third lens, and R6 represents a central curvature radius of an image-side surface of the third lens.

[0006] As an improvement, the camera optical lens further satisfies a following relationship: $1.00 \leq d6/d8 \leq 3.00$. Where, d6 represents an on-axis distance from the image-side surface of the third lens to an object-side surface of the fourth lens, and d8 represents an on-axis distance from an image-side surface of the fourth lens to the object-side surface of the fifth lens.

[0007] As an improvement, the camera optical lens further satisfies a following relationship: $-4.00 \leq f3/f \leq -1.20$; where, f3 represents a focal length of the third lens.

[0008] As an improvement, an object-side surface of the first lens is convex in a paraxial region and an image-side surface of the first lens is concave in the paraxial region. The camera optical lens further satisfies a following relationship: $0.68 \leq f1/f \leq 2.31$; $-3.87 \leq (R1+R2)/(R1-R2) \leq -1.13$; and $0.04 \leq d1/TTL \leq 0.25$; where f1 represents a focal length of the first lens, R1 represents a central curvature radius of the object-side surface of the first lens, R2 represents a central curvature radius of the image-side surface of the first lens, and TTL represents a total track length of the camera optical lens.

[0009] As an improvement, the object-side surface of the second lens is convex in a paraxial region. The camera optical lens further satisfies a following relationship: $-2.65 \leq (R3+R4)/(R3-R4) \leq -0.38$; and $0.04 \leq d3/TTL \leq 0.15$; where R3 represents a central curvature radius of the object-side surface of the second lens, R4 represents a central curvature radius of an image-side surface of the second lens, d3 represents an on-axis thickness of the second lens, and TTL represents a total track length of the camera optical lens.

[0010] As an improvement, the object-side surface of the third lens is concave in a paraxial region and the image-side surface of the third lens is convex in the paraxial region. The camera optical lens further satisfies a following relationship: $-5.98 \leq (R5+R6)/(R5-R6) \leq -0.76$; and $0.03 \leq d5/TTL \leq 0.12$; where d5 represents an on-axis thickness of the third lens and TTL represents a total track length of the camera optical lens.

[0011] As an improvement, an image-side surface of the fourth lens is convex in a paraxial region. The camera optical lens further satisfies a following relationship: $0.28 \leq f4/f \leq 1.32$; $0.43 \leq (R7+R8)/(R7-R8) \leq 2.45$; and $0.05 \leq d7/TTL \leq 0.24$; where f4 represents a focal length of the fourth lens, R7 represents a central curvature radius of an object-side surface of the fourth lens, R8 represents a central curvature radius of the image-side surface of the fourth lens, d7 represents an on-axis thickness of the fourth lens, and TTL represents a total track length of the camera optical lens.

[0012] As an improvement, the object-side surface of the fifth lens is convex in a paraxial region and the image-side surface of the fifth lens is concave in the paraxial region. The camera optical lens further satisfies a following relationship: $-2.42 \leq f5/f \leq -0.39$; and $0.07 \leq d9/TTL \leq 0.29$; where f5 represents a focal length of the fifth lens, d9 represents an on-axis thickness of the fifth lens and TTL represents a total track length of the camera optical lens.

[0013] As an improvement, the camera optical lens further satisfies a following relationship: $TTL/IH \leq 1.40$; where TTL represents a total track length of the camera optical lens and IH represents a maximum image height of the camera optical lens.

[0014] As an improvement, the camera optical lens further satisfies a following relationship: $0.50 \leq f12/f \leq 1.94$; where f12 represents a combined focal length of the first lens and the second lens.

[0015] The present disclosure has the beneficial effects in that: the camera optical lens according to the present disclosure has good optical performance, has the characteristics of large aperture, wide-angle and ultra-thinness and is particularly suitable for mobile phone camera lens assemblies and WEB camera lenses composed of camera elements such as charge coupled device (CCD) or complementary metal-oxide semiconductor sensor (CMOS sensor) for high pixels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] In order to more clearly illustrate the technical solutions in the embodiments of the present disclosure, the accompanying drawings to be used in the description of the embodiments will be briefly described below, and it is obvious that the accompanying drawings in the following description are only some of the embodiments of the present disclosure, and that, for a person of ordinary skill in the art, it is possible to obtain other drawings based on these drawings without any creative effort.

[0017] FIG. 1 is a schematic diagram of a camera optical lens in accordance with a first embodiment of the present disclosure.

[0018] FIG. 2 shows the longitudinal aberration of the camera optical lens shown in FIG. 1.

[0019] FIG. 3 shows the lateral color of the camera optical lens shown in FIG. 1.

[0020] FIG. 4 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 1.

[0021] FIG. 5 is a schematic diagram of a camera optical lens in accordance with a second embodiment of the present disclosure.

[0022] FIG. 6 shows the longitudinal aberration of the camera optical lens shown in FIG. 5.

[0023] FIG. 7 shows the lateral color of the camera optical lens shown in FIG. 5.

[0024] FIG. 8 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 5.

[0025] FIG. 9 is a schematic diagram of a camera optical lens in accordance with a third embodiment of the present disclosure.

[0026] FIG. 10 shows the longitudinal aberration of the camera optical lens shown in FIG. 9.

[0027] FIG. 11 shows the lateral color of the camera optical lens shown in FIG. 9.

[0028] FIG. 12 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 9.

[0029] FIG. 13 is a schematic diagram of a camera optical lens in accordance with a fourth embodiment of the present disclosure.

[0030] FIG. 14 shows the longitudinal aberration of the camera optical lens shown in FIG. 13.

[0031] FIG. 15 shows the lateral color of the camera optical lens shown in FIG. 13.

[0032] FIG. 16 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 13.

[0033] FIG. 17 is a schematic diagram of a camera optical lens in accordance with a comparative example.

[0034] FIG. 18 shows the longitudinal aberration of the camera optical lens shown in FIG. 17.

[0035] FIG. 19 shows the lateral color of the camera optical lens shown in FIG. 17.

[0036] FIG. 20 presents a schematic diagram of the field curvature and distortion of the camera optical lens shown in FIG. 17.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0037] In order to make the objectives, technical solutions and advantages of the present disclosure clearer, various embodiments of the present disclosure will be described in detail below in connection with the accompanying drawings. However, a person of ordinary skill in the art should understand that in the various embodiments of the present disclosure, a number of technical details have been proposed in order to enable the reader to better understand the present disclosure. However, even without these technical details and various variations and modifications based on the following embodiments, the technical solutions claimed to be protected by the present disclosure can be realized.

[0038] As referring to the figures, the present disclosure provides a camera optical lens 10, 20, 30 and 40. FIG. 1, FIG. 5, FIG. 9 and FIG. 13 show the camera optical lens 10, 20, 30 and 40 of the present disclosure, respectively. The camera optical lens 10, 20, 30 and 40 include 5 lenses, respectively. Specifically, from the object side to the image side, the camera optical lens includes in sequence: an aperture S1, a first lens L1, a second lens L2, a third lens L3, a fourth lens L4, and a fifth lens L5. Optical element like optical filter GF can be arranged between the fifth lens L5 and the image surface Si.

[0039] The first lens L1 is made of plastic material, the second lens L2 is made of plastic material, the third lens L3 is made of plastic material, the

fourth lens L4 is made of plastic material, and the fifth lens L5 is made of plastic material. Each lens may also be made of other materials.

[0040] Here, the focal length of the whole camera optical lens is defined as f , the focal length of the second lens L2 is defined as f_2 . The camera optical lens further satisfies the following relationship formula: $3.00 \leq f_2/f \leq 12.00$, by which, a ratio of the focal length of the second lens L2 and the focal length f of the whole camera optical lens is specified, and within the range of the relationship formula, the camera optical lens is made to have better imaging quality and lower sensitivity by a reasonable distribution of the optical focal length of the camera optical lens.

[0041] The curvature radius of the object-side surface of the fifth lens L5 is defined as R_9 , the curvature radius of the image-side surface of the fifth lens L5 is defined as R_{10} . The following relationship formula should be satisfied: $1.10 \leq (R_9 + R_{10}) / (R_9 - R_{10}) \leq 1.90$, by which, the shape of the fifth lens L5 is fixed, and it is conducive to correcting the astigmatism and distortion of the camera optical lens, to enable the distortion to satisfy $|\text{Distortion}| \leq 2.7\%$, such that the generation of dark corners can be reduced.

[0042] The on-axis thickness of the first lens L1 is defined as d_1 , and the on-axis distance from the image-side surface of the first lens L1 to the object-side surface of the second lens L2 is defined as d_2 . The following relationship formula should be satisfied: $1.00 \leq d_1/d_2 \leq 4.00$, by which, a ratio of the on-axis thickness d_1 of the first lens L1 and the on-axis distance d_2 from the image-side surface of the first lens L1 to the object-side surface of the second lens L2 is specified. Within the range of the relationship formula, it is conducive to reducing the total optical length of the whole camera optical lens.

[0043] The curvature radius of the object-side surface of the third lens L3 is defined as R_5 , the curvature radius of the image-side surface of the third lens L3 is defined as R_6 . The following relationship formula should be satisfied: $2.00 \leq R_6/R_5 \leq 15.00$, which is beneficial for the shaping of the third lens L3. Within the range of the relationship formula, a degree of deflection of light passing through the lens can be alleviated, and aberrations can be effectively corrected, thus enabling the chromatic aberration to satisfy $|\text{LC}| \leq 5.0 \mu\text{m}$.

[0044] When the above relationship formulas are satisfied, each of the camera optical lenses **10**, **20**, **30**, and **40** has good optical performance and can meet the design requirement of large aperture, wide-angle and ultra-thinness. According to the characteristics of the camera optical lenses **10**, **20**, **30** and **40**, the camera optical lenses **10**, **20**, **30**, **40**, and **50** are particularly suitable for mobile phone camera lens assemblies and WEB camera lenses composed of camera elements such as charge coupled device (CCD) or complementary metal-oxide semiconductor sensor (CMOS sensor) for high pixels.

[0045] Based on the above relationship formulas and the functions that can be realized, the characteristics of each lens of the lenses are further refined as follows.

[0046] The on-axis distance from the image-side surface of the third lens L3 to the object-side surface of the fourth lens L4 is defined as d_6 , and the on-axis distance from the image-side surface of the fourth lens L4 to the object-side surface of the fifth lens L5 is defined as d_8 . The following relationship formula should be satisfied: $1.00 \leq d_6/d_8 \leq 3.00$, by which, a ratio of the air spacing between the third lens L3 and the fourth lens L4 and the air spacing between the fourth lens L4 and the fifth lens L5 is specified. Within the range of the relationship formula, it is conducive to reducing the total optical length of the whole camera optical lens.

[0047] The focal length of the third lens L3 is defined as f_3 , satisfying the following relationship formula: $-4.00 \leq f_3/f \leq -1.20$, by which, a ratio of the focal length of the third lens L3 and the focal length f of the whole camera optical lens is specified, and within the range of the relationship formula, the camera optical lens is made to have better imaging quality and lower sensitivity by reasonably distributing the optical focal length of the camera optical lens.

[0048] An object-side surface of the first lens L1 is convex in a paraxial region and the image-side surface of the first lens L1 is concave in the paraxial region, and the first lens L1 has a positive refractive power. The object-side surface and image-side surface of the first lens L1 may also be provided in other concave and convex distributions.

[0049] The focal length of the first lens L1 is defined as f_1 , satisfying the following relationship formula: $0.68 \leq f_1/f \leq 2.31$, by which, a ratio of the focal length of the first lens L1 and the focal length f of the whole camera optical lens is specified. Within the specified range, the first lens L1 has an appropriate positive refractive power, which is conducive to reducing systematic aberration, and at the same time is conducive to the development of the lens to ultra-thin and wide-angle.

[0050] The curvature radius of the object-side surface of the first lens L1 is defined as R_1 , the curvature radius of the image-side surface of the first lens L1 is defined as R_2 . The following relationship formula should be satisfied: $-3.87 \leq (R_1 + R_2) / (R_1 - R_2) \leq -1.13$. By reasonably controlling the shape of the first lens L1, the first lens L1 can effectively correct the system spherical aberration. Preferably, the relationship formula $-2.42 \leq (R_1 + R_2) / (R_1 - R_2) \leq -1.41$ shall be satisfied.

[0051] The on-axis thickness of the first lens L1 is defined as d_1 , and the total track length of the camera optical lens is defined as TTL. The following relationship formula: $0.04 \leq d_1/\text{TTL} \leq 0.25$ should be satisfied. Within the range of the relationship formula, it is beneficial for realization of the ultra-thin lens. Preferably, the relationship formula $0.07 \leq d_1/\text{TTL} \leq 0.20$ shall be satisfied.

[0052] The object-side surface of the second lens L2 is convex in a paraxial region and an image-side surface of the second lens L2 is concave or convex in the paraxial region, and the second lens L2 has a positive refractive power. The object-side surface of the second lens L2 may also be provided in other concave and convex distributions.

[0053] The curvature radius of the object-side surface of the second lens L2 is defined as R_3 , the curvature radius of the image-side surface of the second lens L2 is defined as R_4 . The following relationship formula should be satisfied: $-2.65 \leq (R_3 + R_4) / (R_3 - R_4) \leq -0.38$, which specifies the shape of the second lens L2. Within the range of the relationship formula, it is beneficial to correct the problem of on-axis color aberration with the development of lenses to ultra-thin and wide-angle. Preferably, the relationship formula $-1.65 \leq (R_3 + R_4) / (R_3 - R_4) \leq -0.47$ shall be satisfied.

[0054] The on-axis thickness of the second lens L2 is defined as d_3 . The following relationship formula: $0.04 \leq d_3/\text{TTL} \leq 0.15$ should be satisfied. Within the range of the relationship formula, it is beneficial for realization of the ultra-thin lens. Preferably, the relationship formula $0.06 \leq d_3/\text{TTL} \leq 0.12$ shall be satisfied.

[0055] The object-side surface of the third lens L3 is concave in a paraxial region and the image-side surface of the third lens L3 is convex in the paraxial region, and the third lens L3 has a negative refractive power. The object and image-side surfaces of the third lens L3 may also be provided in other concave and convex distributions.

[0056] The camera optical lens may further satisfy the following relationship formula: $-5.98 \leq (R_5 + R_6) / (R_5 - R_6) \leq -0.76$, by which, the shape of the third lens L3 is specified, and it is beneficial for the shaping of the third lens L3. Within the range of the relationship formula, a degree of deflection of light passing through the lens can be alleviated, and aberrations can be effectively corrected. Preferably, the relationship formula of $-3.74 \leq (R_5 + R_6) / (R_5 - R_6) \leq -0.95$ shall be satisfied.

[0057] The on-axis thickness of the third lens L3 is defined as d_5 . The following relationship formula: $0.03 \leq d_5/\text{TTL} \leq 0.12$ should be satisfied. Within the range of the relationship formula, it is beneficial for realization of the ultra-thin lens. Preferably, the relationship formula $0.05 \leq d_5/\text{TTL} \leq 0.10$ shall be satisfied.

[0058] The object-side surface of the fourth lens L4 is concave or convex in a paraxial region and the image-side surface of the fourth lens L4 is convex in the paraxial region, and the fourth lens L4 has a positive refractive power. The image-side surface of the fourth lens L4 may also be provided in other concave and convex distributions.

[0059] The focal length of the fourth lens L4 is f_4 . The following relationship formula should be satisfied: $0.28 \leq f_4/f \leq 1.32$. The system is made to have better imaging quality and lower sensitivity by reasonably distributing the optical focal length of the camera optical lens. Preferably, the relationship formula $0.44 \leq f_4/f \leq 1.06$ shall be satisfied.

[0060] The curvature radius of the object-side surface of the fourth lens L4 is defined as R_7 , the curvature radius of the image-side surface of the

fourth lens **L4** is defined as $R8$. The following relationship formula should be satisfied: $0.43 \leq (R7+R8)/(R7-R8) \leq 2.45$, by which, the shape of the fourth lens **L4** is specified. Within the range of the relationship formula, it is conducive to correcting a problem of an off-axis aberration with the development into the direction of ultra-thin and wide-angle lenses. Preferably, the following relationship formula shall be satisfied, $0.68 \leq (R7+R8)/(R7-R8) \leq 1.96$.

[0061] The on-axis thickness of the fourth lens **L4** is defined as $d7$. The following relationship formula: $0.05 \leq d7/TTL \leq 0.24$ should be satisfied. With the range of the relationship formula, it is beneficial for realization of the ultra-thin lens. Preferably, the relationship formula $0.09 \leq d7/TTL \leq 0.19$ shall be satisfied.

[0062] The object-side surface of the fifth lens **L5** is convex in a paraxial region and the image-side surface of the fifth lens **L5** is concave in the paraxial region, and the fifth lens **L5** has a negative refractive power. The object-side surface and image-side surface of the fifth lens **L5** may also be provided in other concave and convex distributions.

[0063] The focal length of the fifth lens **L5** is $f5$. The following relationship formula should be satisfied: $-2.42 \leq f5/f \leq -0.39$, which can effectively smooth the light angles of the camera optical lens and reduce the tolerance sensitivity. Preferably, the relationship formula $-1.51 \leq f5/f \leq -0.49$ should be satisfied.

[0064] The on-axis thickness of the fifth lens **L5** is defined as $d9$. The following relationship formula: $0.07 \leq d9/TTL \leq 0.29$ should be satisfied. Within the range of the relationship formula, it is beneficial for realization of the ultra-thin lens. Preferably, the relationship formula $0.10 \leq d9/TTL \leq 0.23$ shall be satisfied.

[0065] The maximum image height of the camera optical lens is defined as IH , which satisfies the following relationship formula: $TTL/IH \leq 1.40$.

[0066] The combined focal length of the first lens **L1** and the second lens **L2** is defined as $f12$, which satisfies the following relationship formula: $0.50 \leq f12/f \leq 1.94$. Within the range of the relationship formula, it is beneficial for avoiding the aberration and distortion of the camera optical lens, suppressing the rear focal length, and maintaining the miniaturization of image lens system groups. Preferably, the relationship formula $0.80 \leq f12/f \leq 1.56$ should be satisfied.

[0067] The field of view (FOV) of the camera optical lens is greater than or equal to 90.00° , thereby realizing wide-angle.

[0068] The aperture value (focal number, FNO) of the camera optical lens is less than or equal to 2.11, thereby realizing a large aperture, such that the camera optical lens has good imaging performance.

[0069] The camera optical lens of the present disclosure will be explained with specific embodiments illustrated below. The symbols described in each embodiment are shown below. The unit of the focal length, the on-axis distance, the central curvature radius, the on-axis thickness, inflexion point positions, and arrest point positions is mm.

[0070] TTL: total track length (the on-axis distance from the object-side surface of the first lens **L1** to the image surface S_i), and unit of the TTL being in mm.

[0071] Aperture value (focal number, FNO): a ratio of the effective focal length of the camera optical lens to the diameter of the pupil.

[0072] The technical proposal of the present disclosure will be described in detail in four embodiments, and a comparative example will be provided as a reference for illustration. The technical effect of the present disclosure could not be achieved when value of the parameters exceeds beyond the range of the above-mentioned conditional formula.

First Embodiment

[0073] Table 1 and table 2 show design data of the camera optical lens **10** according to the first embodiment of the present disclosure.

TABLE-US-00001 TABLE 1 R d nd vd $S1 \infty$ $d0 = -0.147$ R1 1.677 $d1 = 0.586$ nd1 1.5444 v1 55.82 R2 5.467 $d2 = 0.283$ R3 13.018 $d3 = 0.325$ nd2 1.5444 v2 55.82 R4 -51.907 $d4 = 0.139$ R5 -3.661 $d5 = 0.295$ nd3 1.6700 v3 19.39 R6 -18.147 $d6 = 0.192$ R7 -44.437 $d7 = 0.672$ nd4 1.5444 v4 55.82 R8 -0.953 $d8 = 0.115$ R9 7.875 $d9 = 0.562$ nd5 1.5444 v5 55.82 R10 0.866 $d10 = 0.629$ R11 ∞ $d11 = 0.210$ ndg 1.5168 vg 64.17 R12 ∞ $d12 = 0.293$

[0074] The meaning of the various symbols is as follows. [0075] $S1$: aperture; [0076] R: the curvature radius at the center of the optical surface;

[0077] R1: the central curvature radius of the object-side surface of the first lens **L1**; [0078] R2: the central curvature radius of the image-side surface of the first lens **L1**; [0079] R3: the central curvature radius of the object-side surface of the second lens **L2**; [0080] R4: the central curvature radius of the image-side surface of the second lens **L2**; [0081] R5: the central curvature radius of the object-side surface of the third lens **L3**; [0082] R6: the central curvature radius of the image-side surface of the third lens **L3**; [0083] R7: the central curvature radius of the object-side surface of the fourth lens **L4**; [0084] R8: the central curvature radius of the image-side surface of the fourth lens **L4**; [0085] R9: the central curvature radius of the object-side surface of the fifth lens **L5**; [0086] R10: the central curvature radius of the image-side surface of the fifth lens **L5**; [0087] R11: the central curvature radius of the object-side surface of the optical filter GF; [0088] R12: the central curvature radius of the image-side surface of the optical filter GF; [0089] d: the on-axis thickness of the lens and the on-axis distance between the lenses; [0090] $d0$: the on-axis distance from the aperture $S1$ to the object-side surface of the first lens **L1**; [0091] $d1$: the on-axis thickness of the first lens **L1**; [0092] $d2$: the on-axis distance from the image-side surface of the first lens **L1** to the object-side surface of the second lens **L2**; [0093] $d3$: the on-axis thickness of the second lens **L2**; [0094] $d4$: the on-axis distance from the image-side surface of the second lens **L2** to the object-side surface of the third lens **L3**; [0095] $d5$: the on-axis thickness of the third lens **L3**; [0096] $d6$: the on-axis distance from the image-side surface of the third lens **L3** to the object-side surface of the fourth lens **L4**; [0097] $d7$: the on-axis thickness of the fourth lens **L4**; [0098] $d8$: the on-axis distance from the image-side surface of the fourth lens **L4** to the object-side surface of the fifth lens **L5**; [0099] $d9$: the on-axis thickness of the fifth lens **L5**; [0100] $d10$: the on-axis distance from the image-side surface of the fifth lens **L5** to the object-side surface of the optical filter GF; [0101] $d11$: the on-axis thickness of the optical filter GF; [0102] $d12$: the on-axis distance from the image-side surface of the optical filter GF to the image surface S_i ; [0103] nd: the refractive index of the d line (the d line refers to green light with a wavelength of 550 nm); [0104] nd1: the refractive index of the d line of the first lens **L1**; [0105] nd2: the refractive index of the d line of the second lens **L2**; [0106] nd3: the refractive index of the d line of the third lens **L3**; [0107] nd4: the refractive index of the d line of the fourth lens **L4**; [0108] nd5: the refractive index of the d line of the fifth lens **L5**; [0109] ndg: the refractive index of the d line of the optical filter GF; [0110] vd: the abbe number; [0111] v1: the abbe number of first lens **L1**; [0112] v2: the abbe number of second lens **L2**; [0113] v3: the abbe number of the third lens **L3**; [0114] v4: the abbe number of the fourth lens **L4**; [0115] v5: the abbe number of the fifth lens **L5**; and [0116] vg: the abbe number of the optical filter GF.

[0117] Table 2 shows aspherical data of each of the lenses in the camera optical lens **10** according to the first embodiment of the present disclosure.

TABLE-US-00002 TABLE 2 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -4.8018E+00 1.7612E-01 -6.5923E-01 5.6619E+00 -6.7759E+01 6.2999E+02 R2 -4.2656E+01 2.8258E-01 -8.4496E+00 1.3640E+02 -1.4158E+03 9.6912E+03 R3 -9.9000E+01 2.3363E-01 -1.2070E+01 2.1379E+02 -2.3860E+03 1.7755E+04 R4 8.9297E+01 -4.6730E-01 3.8644E+00 -4.8494E+01 4.1889E+02 -2.6434E+03 R5 7.6330E+00 -1.2550E+00 2.3096E+01 -3.8264E+02 4.0529E+03 -2.9209E+04 R6 -9.8530E+01 -2.3078E-01 5.9433E-01 -3.7885E+00 -4.4505E+01 5.7497E+02 R7 -9.2030E+01 3.7511E-01 1.1913E-01 -1.0146E+01 5.3498E+01 -1.6270E+02 R8 -9.3454E-01 8.9945E-01 -1.8662E+00 1.4199E+00 4.1130E+00 -1.6352E+01 R9 9.2941E+00 3.4792E-01 -2.2898E+00 4.9598E+00 -6.9001E+00 6.7422E+00 R10 -2.9450E+00 -3.0064E-01 2.8607E-01 -2.1384E-01 1.3334E-01 -7.4254E-02 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 A22 R1 -4.8018E+00 -3.6958E+03 1.3562E+04 -3.1248E+04 4.3997E+04 -3.4624E+04 R2 -4.2656E+01 -4.4675E+04 1.3921E+05 2.8868E+05 3.8112E+05 2.8951E+05 R3 -9.9000E+01 -9.0942E+04 3.2424E+05 -8.0167E+05 1.3456E+06 -1.4611E+06 R4 8.9297E+01 1.2351E+04 -4.2383E+04 1.0501E+05 -1.8390E+05 2.2074E+05 R5 7.6330E+00 1.4863E+05 -5.4423E+05 1.4462E+06 -2.7864E+06 3.8481E+06 R6 -9.8530E+01 -2.9881E+03 9.3230E+03 -1.9375E+04 2.7847E+04 -2.7878E+04 R7 -9.2030E+01 3.3800E+02 -5.0559E+02 5.5511E+02 -4.4807E+02 2.6245E+02 R8 -9.3454E-01 2.9670E+01 -3.4579E+01 2.8161E+01 -1.6504E+01

6.9677E+00 R9 9.2941E+00 -4.7124E+00 2.3779E+00 -8.7117E-01 2.3149E-01 -4.4131E-02 R10 -2.9450E+00 3.6714E-02 -1.4988E-02 4.7069E-03 -1.0885E-03 1.7986E-04 Conic coefficient Aspheric surface coefficients k A24 A26 A28 A30 R1 -4.8018E+00 1.1678E+04 0.0000E+00 0.0000E+00 0.0000E+00 R2 -4.2656E+01 9.6225E+04 0.0000E+00 0.0000E+00 0.0000E+00 R3 -9.9000E+01 9.2495E+05 -2.5905E+05 0.0000E+00 0.0000E+00 R4 8.9297E+01 -1.7232E+05 7.8729E+04 -1.5978E+04 0.0000E+00 R5 7.6330E+00 -3.7089E+06 2.3675E+06 -8.9907E+05 1.5372E+05 R6 -9.8530E+01 1.9122E+04 -8.5772E+03 2.2679E+03 -2.6812E+02 R7 -9.2030E+01 -1.0824E+02 2.9694E+01 -4.8422E+00 3.5328E-01 R8 -9.3454E-01 -2.0708E+00 4.1080E-01 -4.8734E-02 2.6095E-03 R9 9.2941E+00 5.8807E-03 -5.1999E-04 2.7412E-05 -6.5203E-07 R10 -2.9450E+00 -2.0533E-05 1.5336E-06 -6.7287E-08 1.3132E-09

[0118] For convenience, the aspherical surface of each lens surface is the aspherical surface shown in the following formula (1). However, the present disclosure is not limited to the aspherical polynomial form represented by the formula (1).
[00001]

$$z = (cr^2) / \{1 + [1 - (k + 1)(c^2 r^2)]^{1/2}\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} + A14r^{14} + A16r^{16} + A18r^{18} + A20r^{20} + A22r^{22} + A24r^{24} + A26r^{26} +$$

[0119] Where, k represents the Conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, A20, A22, A24, A26, A28, and A30 represent Aspheric surface coefficients, c represents curvature at the center of the optical surface, r represents the vertical distance between the point on the aspherical curve and the optical axis, and z represents the aspherical depth (the vertical distance between the point on the aspherical surface from the optical axis by r and a tangent plane tangent to the vertex on the optical axis of the aspherical surface).

[0120] FIG. 2 and FIG. 3 show the longitudinal aberration and lateral color schematic diagrams after light with a wavelength of 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm passes the camera optical lens 10 in the first embodiment. FIG. 4 shows the field curvature and distortion schematic diagrams after light with a wavelength of 555 nm passes the camera optical lens 10 in the first embodiment, the field curvature S in FIG. 4 is a field curvature in the sagittal direction, T is a field curvature in the meridian direction.

[0121] In this embodiment, the pupil entering diameter (ENPD) of the camera optical lens 10 is 1.449 mm, the full field-of-view image-height IH is 3.269 mm, and the field of view FOV in the diagonal direction is 92.63°. The camera optical lens 10 meets the design requirements of large aperture, ultra-thin, and wide-angle, and the on-axis and off-axis chromatic aberrations are fully corrected, thus having excellent optical characteristics.

Second Embodiment

[0122] The meaning of symbols of the second embodiment is the same as that of the first embodiment.

[0123] FIG. 5 shows a camera optical lens 20 according to a second embodiment of the present disclosure.

[0124] Table 3 and table 4 show design data of the camera optical lens 20 according to the second embodiment of the present disclosure.

TABLE-US-00003 TABLE 3 R d nd vd S1 ∞ d0 = -0.101 R1 1.648 d1 = 0.380 nd1 1.5444 v1 55.82 R2 5.418 d2 = 0.377 R3 20.411 d3 = 0.330 nd2 1.5444 v2 55.82 R4 -329.428 d4 = 0.108 R5 -3.841 d5 = 0.338 nd3 1.6700 v3 19.39 R6 -7.705 d6 = 0.214 R7 -19.646 d7 = 0.678 nd4 1.5444 v4 55.82 R8 -0.950 d8 = 0.124 R9 7.972 d9 = 0.555 nd5 1.5444 v5 55.82 R10 0.844 d10 = 0.627 R11 ∞ d11 = 0.210 ndg 1.5168 vg 64.17 R12 ∞ d12 = 0.292

[0125] Table 4 shows aspherical data of each lens in the camera optical lens 20 according to the second embodiment of the present disclosure.

TABLE-US-00004 TABLE 4 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -4.3936E+00 1.7704E-01 -6.7159E-01 5.6564E+00 -6.7761E+01 6.2998E+02 R2 -4.8257E+01 2.9549E-01 -8.4245E+00 1.3639E+02 -1.4158E+03 9.6912E+03 R3 4.1595E+02 2.4457E-01 -1.2109E+01 2.1379E+02 -2.3860E+03 1.7756E+04 R4 1.4979E+05 -5.4752E-01 3.8525E+00 -4.8499E+01 4.1886E+02 -2.6434E+03 R5 1.0284E+01 -1.2736E+00 2.3066E+01 -3.8265E+02 4.0529E+03 -2.9209E+04 R6 -4.3046E+02 -2.2699E-01 5.9015E-01 -3.7955E+00 -4.4508E+01 5.7497E+02 R7 -3.1129E+03 3.7658E-01 1.1890E-01 -1.0147E+01 5.3499E+01 -1.6270E+02 R8 -9.4507E-01 9.0210E-01 -1.8652E+00 1.4202E+00 4.1131E+00 -1.6352E+01 R9 9.3316E+00 3.4853E-01 -2.2897E+00 4.9598E+00 -6.9001E+00 6.7422E+00 R10 -2.8484E+00 -3.0045E-01 2.8636E-01 -2.1384E-01 1.3334E-01 -7.4254E-02 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 A22 R1 -4.3936E+00 -3.6958E+03 1.3562E+04 -3.1249E+04 4.3996E+04 -3.4624E+04 R2 -4.8257E+01 -4.4675E+04 1.3921E+05 -2.8868E+05 3.8112E+05 -2.8951E+05 R3 4.1595E+02 -9.0942E+04 3.2424E+05 -8.0167E+05 1.3456E+06 -1.4611E+06 R4 1.4979E+05 1.2351E+04 -4.2383E+04 1.0501E+05 -1.8390E+05 2.2074E+05 R5 1.0284E+01 1.4863E+05 -5.4423E+05 1.4462E+06 -2.7864E+06 3.8481E+06 R6 -4.3046E+02 -2.9881E+03 9.3230E+03 -1.9375E+04 2.7847E+04 -2.7878E+04 R7 -3.1129E+03 3.3800E+02 -5.0559E+02 5.5511E+02 -4.4807E+02 2.6245E+02 R8 -9.4507E-01 2.9671E+01 -3.4579E+01 2.8161E+01 -1.6504E+01 6.9677E+00 R9 9.3316E+00 -4.7124E+00 2.3779E+00 -8.7117E-01 2.3149E-01 -4.4131E-02 R10 -2.8484E+00 3.6714E-02 -1.4988E-02 4.7069E-03 -1.0885E-03 1.7986E-04 Conic coefficient Aspheric surface coefficients k A24 A26 A28 A30 R1 -4.3936E+00 1.1679E+04 0.0000E+00 0.0000E+00 0.0000E+00 R2 -4.8257E+01 9.6231E+04 0.0000E+00 0.0000E+00 0.0000E+00 R3 4.1595E+02 9.2495E+05 -2.5906E+05 0.0000E+00 0.0000E+00 R4 1.4979E+05 -1.7232E+05 7.8729E+04 -1.5978E+04 0.0000E+00 R5 1.0284E+01 -3.7089E+06 2.3675E+06 -8.9907E+05 1.5372E+05 R6 -4.3046E+02 1.9122E+04 -8.5772E+03 2.2679E+03 -2.6812E+02 R7 -3.1129E+03 -1.0824E+02 2.9694E+01 -4.8422E+00 3.5328E-01 R8 -9.4507E-01 -2.0708E+00 4.1080E-01 -4.8734E-02 2.6095E-03 R9 9.3316E+00 5.8807E-03 -5.1999E-04 2.7412E-05 -6.5203E-07 R10 -2.8484E+00 -2.0533E-05 1.5336E-06 -6.7287E-08 1.3132E-09

[0126] FIG. 6 and FIG. 7 show diagrams of longitudinal aberrations and lateral colors after light with a wavelength of 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm passes the camera optical lens 20 of the second embodiment, respectively. FIG. 8 shows a schematic diagram of a field curvature and distortion after light with a wavelength of 555 nm passes the camera optical lens 20 of the second embodiment. The field curvature S in FIG. 8 is a field curvature in the sagittal direction, and T is a field curvature in the meridian direction.

[0127] In this embodiment, the pupil entering diameter ENPD of the camera optical lens 20 is 1.396 mm, the full field-of-view image-height IH is 3.269 mm, and the field of view FOV in the diagonal direction is 93.00°. The camera optical lens 20 meets the design requirements of large aperture, ultra-thin and wide-angle, and the on-axis and off-axis chromatic aberrations are fully corrected, thus having excellent optical characteristics.

Third Embodiment

[0128] The meaning of symbols of the third embodiment is the same as that of the first embodiment.

[0129] FIG. 9 shows a camera optical lens 30 according to a third embodiment of the present disclosure. The object-side surface of the fourth lens L4 is convex in the paraxial region.

[0130] Table 5 and table 6 show design data of the camera optical lens 30 according to the third embodiment of the present disclosure.

TABLE-US-00005 TABLE 5 R d nd vd S1 ∞ d0 = -0.136 R1 1.801 d1 = 0.762 nd1 1.5444 v1 55.82 R2 6.998 d2 = 0.191 R3 6.440 d3 = 0.339 nd2 1.5444 v2 55.82 R4 -23.463 d4 = 0.156 R5 -2.567 d5 = 0.361 nd3 1.6700 v3 19.39 R6 -38.467 d6 = 0.213 R7 12.644 d7 = 0.678 nd4 1.5444 v4 55.82 R8 -0.985 d8 = 0.072 R9 17.916 d9 = 0.665 nd5 1.5444 v5 55.82 R10 0.929 d10 = 0.629 R11 ∞ d11 = 0.210 ndg 1.5168 vg 64.17 R12 ∞ d12 = 0.302

[0131] Table 6 shows aspherical data of each lens in the camera optical lens 30 according to the third embodiment of the present disclosure.

TABLE-US-00006 TABLE 6 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -8.4825E+00 1.8265E-01 -6.2519E-01 5.6201E+00 -6.7879E+01 6.2996E+02 R2 -2.4928E+02 2.5469E-01 -8.4999E+00 1.3630E+02 -1.4158E+03 9.6914E+03 R3 1.1325E+01 2.6269E-01 -1.2176E+01 2.1369E+02 -2.3859E+03 1.7756E+04 R4 7.6405E+02 -4.0950E-01 3.8684E+00 -4.8590E+01 4.1892E+02 -2.6433E+03 R5 7.7557E+00 -1.2381E+00 2.3096E+01 -3.8259E+02 4.0529E+03 -2.9209E+04 R6 9.8507E+02 -2.6603E-01 5.8934E-01 -3.8008E+00 -4.4514E+01 5.7498E+02 R7 -3.2887E+02 3.7445E-01 1.1419E-01 -1.0149E+01 5.3499E+01 -1.6270E+02 R8 -9.3960E-01 9.0510E-01 -1.8689E+00 1.4203E+00 4.1133E+00 -1.6352E+01 R9 3.9278E+01 3.5821E-01 -2.2873E+00 4.9599E+00 -6.9002E+00 6.7422E+00 R10 -2.9951E+00 -2.9410E-01 2.8454E-01 -2.1378E-01 1.3336E-01 -7.4252E-02 Conic coefficient Aspheric surface

coefficients k A14 A16 A18 A20 A22 R1 -8.4825E+00 -3.6956E+03 1.3562E+04 -3.1248E+04 4.3996E+04 -3.4626E+04 R2 -2.4928E+02 -4.4675E+04 1.3921E+05 -2.8868E+05 3.8112E+05 -2.8951E+05 R3 1.1325E+01 -9.0942E+04 3.2424E+05 -8.0167E+05 1.3456E+06 -1.4611E+06 R4 7.6405E+02 1.2351E+04 -4.2383E+04 1.0501E+05 -1.8390E+05 2.2074E+05 R5 7.7557E+00 1.4863E+05 -5.4423E+05 1.4462E+06 -2.7864E+06 3.8481E+06 R6 9.8507E+02 -2.9881E+03 9.3230E+03 -1.9375E+04 2.7847E+04 -2.7878E+04 R7 -3.2887E+02 3.3800E+02 -5.0559E+02 5.5511E+02 -4.4807E+02 2.6245E+02 R8 -9.3960E-01 2.9670E+01 -3.4579E+01 2.8161E+01 -1.6504E+01 6.9677E+00 R9 3.9278E+01 -4.7124E+00 2.3779E+00 -8.7117E-01 2.3149E-01 -4.4131E-02 R10 -2.9951E+00 3.6714E-02 -1.4988E-02 4.7069E-03 -1.0885E-03 1.7986E-04 Conic coefficient Aspheric surface coefficients k A24 A26 A28 A30 R1 -8.4825E+00 1.1678E+04 0.0000E+00 0.0000E+00 0.0000E+00 R2 -2.4928E+02 9.6224E+04 0.0000E+00 0.0000E+00 0.0000E+00 R3 1.1325E+01 9.2495E+05 -2.5905E+05 0.0000E+00 0.0000E+00 R4 7.6405E+02 -1.7232E+05 7.8729E+04 -1.5978E+04 0.0000E+00 R5 7.7557E+00 -3.7089E+06 2.3675E+06 -8.9907E+05 1.5372E+05 R6 9.8507E+02 1.9122E+04 -8.5772E+03 2.2679E+03 -2.6812E+02 R7 -3.2887E+02 -1.0824E+02 2.9693E+01 -4.8421E+00 3.5333E-01 R8 -9.3960E-01 -2.0708E+00 4.1080E-01 -4.8734E-02 2.6094E-03 R9 3.9278E+01 5.8807E-03 -5.1999E-04 2.7412E-05 -6.5202E-07 R10 -2.9951E+00 -2.0533E-05 1.5336E-06 -6.7287E-08 1.3132E-09

[0132] FIG. 10 and FIG. 11 show diagrams of longitudinal aberrations and lateral colors after light with a wavelength of 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm passes the camera optical lens 30 of the third embodiment, respectively. FIG. 12 shows a schematic diagram of a field curvature and distortion after light with a wavelength of 555 nm passes the camera optical lens 30 of the third embodiment. The field curvature S in FIG. 12 is a field curvature in the sagittal direction, and T is a field curvature in the meridian direction.

[0133] In this embodiment, the pupil entering diameter ENPD of the camera optical lens 30 is 1.468 mm, the full field-of-view image-height IH is 3.269 mm, and the field of view FOV in the diagonal direction is 90.01°. The camera optical lens 30 meets the design requirements of large aperture, ultra-thin, and wide-angle, and the on-axis and off-axis chromatic aberrations are fully corrected, thus having excellent optical characteristics.

Fourth Embodiment

[0134] The meaning of symbols of the fourth embodiment is the same as that of the first embodiment.

[0135] FIG. 13 shows a camera optical lens 40 according to a fourth embodiment of the present disclosure. The image-side surface of the second lens L2 is concave in the paraxial region.

[0136] Table 7 and table 8 show design data of the camera optical lens 40 according to the fourth embodiment of the present disclosure.

TABLE-US-00007 TABLE 7 R d nd vd S1 ∞ d0 = -0.071 R1 1.525 d1 = 0.425 nd1 1.5444 v1 55.82 R2 4.794 d2 = 0.311 R3 5.864 d3 = 0.453 nd2 1.5444 v2 55.82 R4 42.189 d4 = 0.087 R5 -2.555 d5 = 0.291 nd3 1.6700 v3 19.39 R6 -10.469 d6 = 0.132 R7 -4.084 d7 = 0.492 nd4 1.5444 v4 55.82 R8 -0.981 d8 = 0.131 R9 3.242 d9 = 0.877 nd5 1.5444 v5 55.82 R10 1.002 d10 = 0.676 R11 ∞ d11 = 0.210 ndg 1.5168 vg 64.17 R12 ∞ d12 = 0.399

[0137] Table 8 shows aspherical data of each lens in the camera optical lens 40 according to the fourth embodiment of the present disclosure.

TABLE-US-00008 TABLE 8 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -5.5609E+00 2.0854E-01 -6.0817E-01 5.5099E+00 -6.7898E+01 6.2999E+02 R2 -1.5413E+00 2.6037E-01 -8.6584E+00 1.3647E+02 -1.4152E+03 9.6919E+03 R3 -6.8843E+01 2.0621E-01 -1.2155E+01 2.1365E+02 -2.3861E+03 1.7755E+04 R4 2.4725E+03 -4.5799E-01 3.6781E+00 -4.8684E+01 4.1893E+02 -2.6432E+03 R5 8.2597E+00 -1.2699E+00 2.3218E+01 -3.8262E+02 4.0528E+03 -2.9209E+04 R6 8.7748E+01 -2.5905E-01 6.0012E-01 -3.7848E+00 -4.4510E+01 5.7497E+02 R7 -3.1805E+02 4.0221E-01 4.5485E-02 -1.0151E+01 5.3506E+01 -1.6270E+02 R8 -8.9934E-01 8.7086E-01 -1.8558E+00 1.4235E+00 4.1137E+00 -1.6352E+01 R9 3.9864E+00 3.1997E-01 -2.2952E+00 4.9513E+00 -6.9010E+00 6.7423E+00 R10 -2.0259E+00 -3.1818E-01 2.8869E-01 -2.1384E-01 1.3332E-01 -7.4257E-02 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 A22 R1 -5.5609E+00 -3.6954E+03 1.3562E+04 -3.1249E+04 4.3993E+04 -3.4632E+04 R2 -1.5413E+00 -4.4677E+04 1.3920E+05 -2.8870E+05 3.8110E+05 -2.8948E+05 R3 1.1325E+01 -9.0943E+04 3.2424E+05 -8.0167E+05 1.3456E+06 -1.4611E+06 R4 2.4725E+03 1.2351E+04 -4.2383E+04 1.0501E+05 -1.8390E+05 2.2074E+05 R5 8.2597E+00 1.4863E+05 -5.4423E+05 1.4462E+06 -2.7864E+06 3.8481E+06 R6 8.7748E+01 -2.9881E+03 9.3230E+03 -1.9375E+04 2.7847E+04 -2.7878E+04 R7 -3.1805E+02 3.3800E+02 -5.0559E+02 5.5511E+02 -4.4807E+02 2.6245E+02 R8 -8.9934E-01 2.9670E+01 -3.4579E+01 2.8161E+01 -1.6504E+01 6.9676E+00 R9 3.9864E+00 -4.7123E+00 2.3779E+00 -8.7121E-01 2.3147E-01 -4.4139E-02 R10 -2.0259E+00 3.6714E-02 -1.4988E-02 4.7069E-03 -1.0885E-03 1.7986E-04 Conic coefficient Aspheric surface coefficients k A24 A26 A28 A30 R1 -5.5609E+00 1.1684E+04 0.0000E+00 0.0000E+00 0.0000E+00 R2 -1.5413E+00 9.6613E+04 0.0000E+00 0.0000E+00 0.0000E+00 R3 -6.8843E+01 9.2498E+05 -2.5902E+05 0.0000E+00 0.0000E+00 R4 2.4725E+03 -1.7232E+05 7.8728E+04 -1.5981E+04 0.0000E+00 R5 8.2597E+00 -3.7089E+06 2.3675E+06 -8.9907E+05 1.5372E+05 R6 8.7748E+01 1.9122E+04 -8.5772E+03 2.2679E+03 -2.6812E+02 R7 -3.1805E+02 -1.0824E+02 2.9694E+01 4.8421E+00 3.5319E-01 R8 -8.9934E-01 -2.0708E+00 4.1080E-01 -4.8729E-02 2.6172E-03 R9 3.9864E+00 5.8771E-03 -5.2122E-04 2.6814E-05 -7.7276E-07 R10 -2.0259E+00 -2.0533E-05 1.5336E-06 -6.7286E-08 1.3136E-09

[0138] FIG. 14 and FIG. 15 show diagrams of longitudinal aberrations and lateral colors after light with a wavelength of 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm passes the camera optical lens 40 of the fourth embodiment, respectively. FIG. 16 shows a schematic diagram of a field curvature and distortion after light with a wavelength of 555 nm passes the camera optical lens 40 of the fourth embodiment. The field curvature S in FIG. 16 is a field curvature in the sagittal direction, and T is a field curvature in the meridian direction.

[0139] In this embodiment, the pupil entering diameter ENPD of the camera optical lens 40 is 1.207 mm, the full field-of-view image-height IH is 3.269 mm, and the field of view FOV in the diagonal direction is 90.88°. The camera optical lens 40 meets the design requirements of large aperture, ultra-thin, and wide-angle, and the on-axis and off-axis chromatic aberrations are fully corrected, thus having excellent optical characteristics.

[0140] Table 11, which appears later, shows the values corresponding to the parameters specified in the conditional formula for each of the values in the first, second, third and fourth embodiments.

Comparative Example

[0141] The meaning of symbols of the comparative example is the same as that of the first embodiment.

[0142] FIG. 17 shows a camera optical lens 50 according to a comparative example.

[0143] Table 9 and table 10 show design data of the camera optical lens 50 of the comparative example.

TABLE-US-00009 TABLE 9 R d nd vd S1 ∞ d0 = -0.107 R1 1.674 d1 = 0.634 nd1 1.5444 v1 55.82 R2 4.282 d2 = 0.247 R3 7.049 d3 = 0.338 nd2 1.5444 v2 55.82 R4 -13.292 d4 = 0.155 R5 -3.196 d5 = 0.317 nd3 1.6700 v3 19.39 R6 -14.948 d6 = 0.212 R7 -22.994 d7 = 0.674 nd4 1.5444 v4 55.82 R8 -0.967 d8 = 0.125 R9 7.900 d9 = 0.552 nd5 1.5444 v5 55.82 R10 0.858 d10 = 0.607 R11 ∞ d11 = 0.210 ndg 1.5168 vg 64.17 R12 ∞ d12 = 0.254

[0144] Table 10 shows aspherical data of each lens in the camera optical lens 50 of the comparative example.

TABLE-US-00010 TABLE 10 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -5.5696E+00 1.6714E-01 -6.2639E-01 5.6588E+00 -6.7798E+01 6.3001E+02 R2 -3.5973E+01 2.8419E-01 -8.4009E+00 1.3627E+02 -1.4158E+03 9.6914E+03 R3 -2.9213E+01 2.5135E-01 -2.0151E+01 2.1357E+02 -2.3859E+03 1.7755E+04 R4 2.3705E+02 -4.4764E-01 3.8446E+00 -4.8469E+01 4.1887E+02 -2.6434E+03 R5 7.7630E+00 -1.2631E+00 2.3109E+01 3.8265E+02 4.0529E+03 -2.9209E+04 R6 -5.0087E+02 -2.3674E-01 5.9631E-01 -3.7931E+00 -4.4511E+01 5.7497E+02 R7 3.0882E+02 3.7827E-01 1.0893E-01 -1.0145E+01 5.3508E+01 -1.6271E+02 R8 -9.0127E-01 8.9272E-01 -1.8619E+00 1.4178E+00 4.1135E+00 -1.6352E+01 R9 9.1665E+00 3.4353E-01 -2.2889E+00 4.9595E+00 -6.9001E+00 6.7422E+00 R10 -2.8549E+00 -2.9979E-01 2.8662E-01 -2.1385E-01 1.3334E-01 -7.4254E-02 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 A22 R1 -5.5696E+00 -3.6958E+03 1.3562E+04 -3.1250E+04 4.3996E+04 -3.4626E+04 R2 -3.5973E+01 -4.4675E+04 1.3921E+05 -2.8868E+05 3.8112E+05 -2.8952E+05 R3 -2.9213E+01 -9.0941E+04 3.2424E+05 -8.0167E+05 1.3456E+06

-1.4611E+06 R4 2.3705E+02 1.2351E+04 -4.2383E+04 1.0501E+05 -1.8390E+05 2.2074E+05 R5 7.7630E+00 1.4863E+05 -5.4423E+05
1.4462E+06 -2.7864E+06 3.8481E+06 R6 -5.0087E+02 -2.9881E+03 9.3230E+03 -1.9375E+04 2.7847E+04 -2.7878E+04 R7 3.0882E+02
3.3799E+02 -5.0559E+02 5.5511E+02 -4.4807E+02 2.6245E+02 R8 -9.0127E-01 2.9670E+01 -3.4579E+01 2.8161E+01 -1.6504E+01
6.9677E+00 R9 9.1665E+00 -4.7124E+00 2.3779E+00 -8.7117E-01 2.3149E-01 -4.4131E-02 R10 -2.8549E+00 3.6714E-02 -1.4988E-02
4.7069E-03 -1.0885E-03 1.7986E-04 Conic coefficient Aspheric surface coefficients k A24 A26 A28 A30 R1 -5.5696E+00 1.1685E+04
0.0000E+00 0.0000E+00 0.0000E+00 R2 -3.5973E+01 9.6236E+04 0.0000E+00 0.0000E+00 0.0000E+00 R3 -2.9213E+01 9.2496E+05
-2.5905E+05 0.0000E+00 0.0000E+00 R4 2.3705E+02 -1.7232E+05 7.8729E+04 -1.5978E+04 0.0000E+00 R5 7.7630E+00 -3.7089E+06
2.3675E+06 8.9907E+05 1.5372E+05 R6 -5.0087E+02 1.9122E+04 -8.5772E+03 2.2679E+03 -2.6812E+02 R7 3.0882E+02 -1.0824E+02
2.9694E+01 -4.8422E+00 3.5329E-01 R8 -9.0127E-01 -2.0708E+00 4.1080E-01 4.8734E-02 2.6095E-03 R9 9.1665E+00 5.8807E-03
-5.1999E-04 2.7412E-05 -6.5205E-07 R10 -2.8549E+00 -2.0533E-05 1.5336E-06 -6.7288E-08 1.3133E-09

[0145] FIG. 18 and FIG. 19 show diagrams of longitudinal aberrations and lateral colors after light with a wavelength of 650 nm, 610 nm, 555 nm, 510 nm, 470 nm, and 435 nm passes the camera optical lens 50 of the comparative example, respectively. FIG. 20 shows a schematic diagram of a field curvature and distortion after light with a wavelength of 555 nm passes the camera optical lens 50 of the comparative example. The field curvature S in FIG. 20 is a field curvature in the sagittal direction, and T is a field curvature in the meridian direction.

[0146] Table 11 below lists the values of the corresponding conditional expressions in the comparative example according to the above conditional expressions. Obviously, the camera optical lens 50 of the comparative example does not satisfy the above-described conditional formula $3.00 \leq f_2 / f \leq 12.00$, thus having poor imaging effect.

[0147] In this embodiment, the pupil entering diameter ENPD of the camera optical lens 50 is 1.444 mm, the full field-of-view image-height IH is 3.269 mm, and the field of view FOV in the diagonal direction is 90.33° , such that the camera optical lens 50 does not meet the design requirements of large aperture, ultra-thin, and wide-angle.

TABLE-US-00011 TABLE 11 Parameter and Embodiment Embodiment Embodiment Embodiment comparative conditional formula 1 2 3 4 example
f2/f 6.255 11.970 3.004 4.882 2.789 (R9 + R10)/(R9 - R10) 1.247 1.237 1.109 1.895 1.244 d1/d2 2.071 1.008 3.990 1.367 2.567 R6/R5 4.957 2.006
14.985 4.097 4.677 f 3.052 2.941 3.092 2.543 3.042 f1 4.201 4.187 4.221 3.915 4.635 f2 19.090 35.203 9.289 12.414 8.483 f3 -6.839 -11.739
-4.085 -5.075 -6.078 f4 1.773 1.804 1.703 2.237 1.828 f5 -1.833 -1.777 -1.819 -3.080 -1.812 f12 3.573 3.813 3.102 3.099 3.230 FNO 2.106
2.107 2.106 2.107 2.107 TTL 4.301 4.233 4.578 4.484 4.325

[0148] Those of ordinary skill in the art will appreciate that the above embodiments are embodiments of the present disclosure, and that in practical application, various changes can be made to them in form and detail without departing from the spirit and scope of the present disclosure.

Claims

1. A camera optical lens, comprising five lenses, wherein the five lenses include, from an object side to an image side in sequence: a first lens having a positive refractive power, a second lens having a positive refractive power, a third lens having a negative refractive power, a fourth lens having a positive refractive power, and a fifth lens having a negative refractive power; wherein the camera optical lens satisfies following relationships:

$$3. \leq f_2 / f \leq 12.;$$

$$1.1 \leq (R_9 + R_{10}) / (R_9 - R_{10}) \leq 1.9;$$

$$1. \leq d_1 / d_2 \leq 4.; \text{ and}$$

$$2. \leq R_6 / R_5 \leq 15.;$$

wherein f represents a focal length of the camera optical lens, f2 represents a focal length of the second lens,

R9 represents a central curvature radius of an object-side surface of the fifth lens, R10 represents a central curvature radius of an image-side surface of the fifth lens, d1 represents an on-axis thickness of the first lens, d2 represents an on-axis distance from an image-side surface of the first lens to an object-side surface of the second lens, R5 represents a central curvature radius of an object-side surface of the third lens, and R6 represents a central curvature radius of an image-side surface of the third lens.

2. The camera optical lens of claim 1, wherein the camera optical lens further satisfies a following relationship: $1. \leq d_6 / d_8 \leq 3.$; wherein d6 represents an on-axis distance from the image-side surface of the third lens to an object-side surface of the fourth lens, and d8 represents an on-axis distance from an image-side surface of the fourth lens to the object-side surface of the fifth lens.

3. The camera optical lens of claim 1, wherein the camera optical lens further satisfies a following relationship: $-4. \leq f_3 / f \leq -1.2$; wherein f3 represents a focal length of the third lens.

4. The camera optical lens of claim 1, wherein an object-side surface of the first lens is convex in a paraxial region and the image-side surface of the first lens is concave in the paraxial region; wherein the camera optical lens further satisfies a following relationship:

$$0.68 \leq f_1 / f \leq 2.31;$$

$$-3.87 \leq (R_1 + R_2) / (R_1 - R_2) \leq -1.13; \text{ and} \quad \text{wherein } f_1 \text{ represents a focal length of the first lens, } R_1 \text{ represents a central curvature radius of the}$$

$$0.04 \leq d_1 / \text{TTL} \leq 0.25;$$

object-side surface of the first lens, R2 represents a central curvature radius of the image-side surface of the first lens, and TTL represents a total track length of the camera optical lens.

5. The camera optical lens of claim 1, wherein the object-side surface of the second lens is convex in a paraxial region; wherein the camera optical lens further satisfies a following relationship: $-2.65 \leq (R_3 + R_4) / (R_3 - R_4) \leq -0.38$; and wherein R3 represents a central curvature radius of the

$$0.04 \leq d_3 / \text{TTL} \leq 0.15;$$

object-side surface of the second lens, R4 represents a central curvature radius of an image-side surface of the second lens, d3 represents an on-axis thickness of the second lens, and TTL represents a total track length of the camera optical lens.

6. The camera optical lens of claim 1, wherein the object-side surface of the third lens is concave in a paraxial region and the image-side surface of the third lens is convex in the paraxial region; wherein the camera optical lens further satisfies a following relationship:

$$-5.98 \leq (R_5 + R_6) / (R_5 - R_6) \leq -0.76; \text{ and}$$

$$0.003 \leq d_5 / \text{TTL} \leq 0.12;$$

wherein d5 represents an on-axis thickness of the third lens and TTL represents a total track length of

the camera optical lens.

7. The camera optical lens of claim 1, wherein an image-side surface of the fourth lens is convex in a paraxial region; wherein the camera optical lens

$$0.28 \leq f_4 / f \leq 1.32;$$

further satisfies a following relationship: $0.43 \leq (R_7 + R_8) / (R_7 - R_8) \leq 2.45$; and wherein f4 represents a focal length of the fourth lens, R7

$$0.05 \leq d_7 / \text{TTL} \leq 0.24;$$

represents a central curvature radius of an object-side surface of the fourth lens, R8 represents a central curvature radius of an image-side surface of the fourth lens, d7 represents an on-axis thickness of the fourth lens, and TTL represents a total track length of the camera optical lens.

8. The camera optical lens of claim 1, wherein the object-side surface of the fifth lens is convex in a paraxial region and the image-side surface of the

fifth lens is concave in the paraxial region; wherein the camera optical lens further satisfies a following relationship: $-2.42 \leq f_5 / f \leq -0.39$; and $0.07 \leq d_9 / \text{TTL} \leq 0.29$;

wherein f_5 represents a focal length of the fifth lens, d_9 represents an on-axis thickness of the fifth lens and TTL represents a total track length of the camera optical lens.

9. The camera optical lens of claim 1, wherein the camera optical lens further satisfies a following relationship: $\text{TTL} / H \leq 1.4$; wherein TTL represents a total track length of the camera optical lens and IH represents a maximum image height of the camera optical lens.

10. The camera optical lens of claim 1, wherein the camera optical lens further satisfies a following relationship: $0.5 \leq f_{12} / f \leq 1.94$; wherein f_{12} represents a combined focal length of the first lens and the second lens.
